The Matter That Never Made a Ruler

From dimensional remainders to dark-sector free fall

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Most of the gravitating matter used to infer the cosmic scale H has supplied none of the clocks, rods or test bodies with which that scale is measured. Standards made from familiar matter are carried into a description of a universe with a larger material inventory. Dimensional analysis separates the chosen units from the physical relations, then leaves exposed the further question: what warrants the belief that every kind of matter realises those standards, and falls with them, in the same way?

Historical observation

Maxwell chose M, L, T in 1873. Buckingham (1914) and Bridgman (1922) formalised the method after Einstein, and kept the same base. The choice is pre-relativistic; the formalism that fixed it is not. [1, 2, 3]

The named-dimension notation used in the companion pieces writes, for example, [velocity] $= LT^{-1}$. Square brackets remove the numerical unit while retaining the kind of quantity. A dimensionally homogeneous equation survives the replacement of metres by feet and seconds by years. This is a severe constraint, but not a dynamical law: it excludes impossible combinations without selecting the ingredients nature uses.

1 Uzan's fork in the road

Jean-Philippe Uzan places the familiar construction of natural units inside a wider metrological argument. A theory supplies dimensioned constants. Three independent dimensions permit three of them to define three fundamental units; what cannot be absorbed into that choice remains as dimensionless physics. Figure 1 reproduces the logic rather than the artwork of his diagram.[4]

For $\hbar, c, G$, inversion of the dimensional relations gives

$$\ell_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^3}}, \qquad m_{\mathrm{P}} = \sqrt{\frac{\hbar c}{G}}, \qquad t_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^5}}.$$

These formulae identify the only monomial length, mass and time available from the nominated ingredients, apart from dimensionless factors. They do not establish that ℓ_{P} is a smallest length, nor supply the 2π that distinguishes h from $\hbar$. The physical premise enters before the algebra, in the choice of ingredients, and again afterwards, in the interpretation.

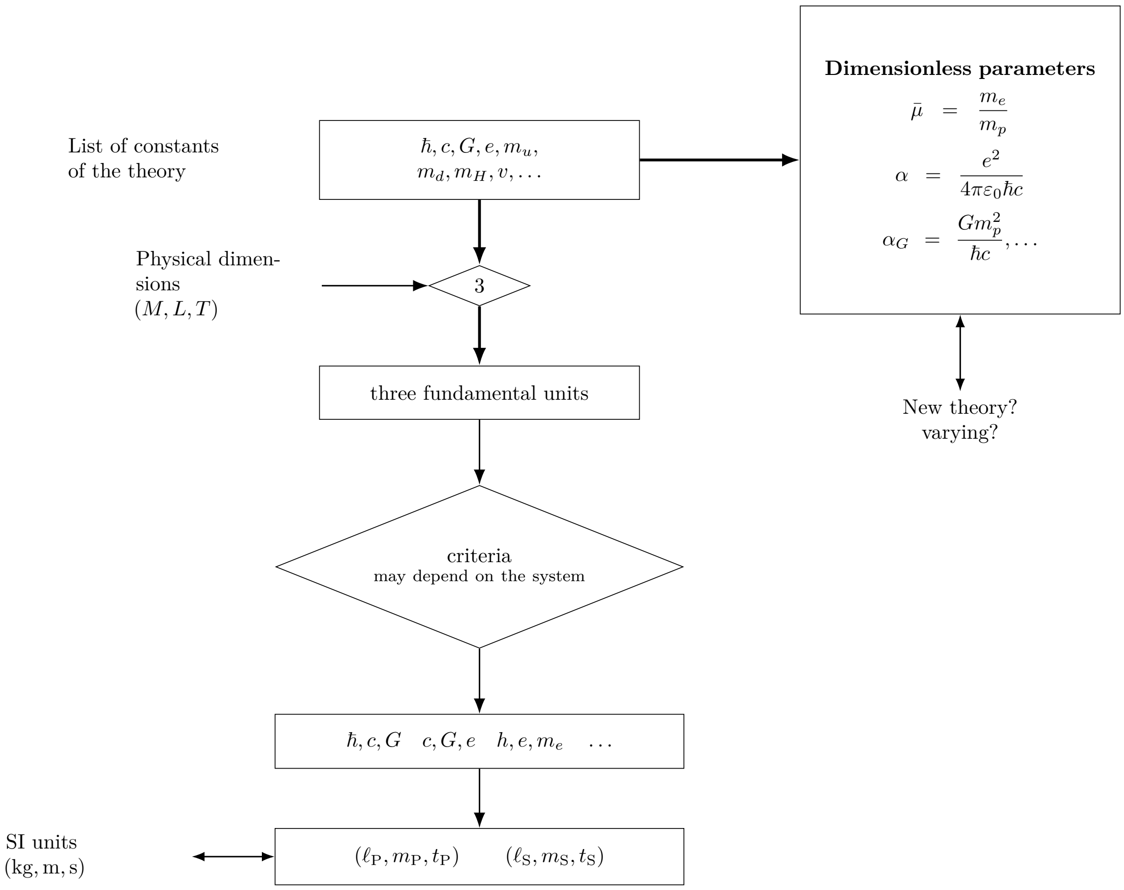

Figure 1: From a theory's dimensioned constants to fundamental units, with dimensionless relations left over. Reproduced in TikZ after Uzan. The boxes in the central construction share one vertical axis.

2 The fourth constant kills the table

Now add the Hubble parameter, $[H] = T^{-1}$. With columns ordered as $(\hbar, c, G, H)$, their powers of M, L, T form the dimensional matrix

$$D = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 2 & 1 & 3 & 0 \\ -1 & -1 & -2 & -1 \end{pmatrix}.$$

Its rank is three and its null space is generated by

$$\boldsymbol{n} = (1, -5, 1, 2)^{\mathsf{T}}, \qquad D\boldsymbol{n} = 0.$$

Thus one convenient generator of the dimensionless remainder is

$$\boxed{\Xi_H \equiv \frac{\hbar G H^2}{c^5}}.$$

We shall call this chosen representative the *Buckingham–Bridgman constant* for the four-quantity set. The name fixes a convenient generator, since Buckingham's result licenses any non-singular reparameterisation of the same information: Ξ_H^{-1}, $\Xi_H^{1/2}$, or more generally $f(\Xi_H)$. "Constant" here first means invariant under passive changes in the sizes of the base units. Independence from cosmic time and invariance under a change of field variables require further physical argument.

Before using that remainder, display the finite menu obtained by choosing the four constants three at a time:

Constants	Length	Mass	Time
$\hbar, c, G$	$\sqrt{\hbar G/c^3}$	$\sqrt{\hbar c/G}$	$\sqrt{\hbar G/c^5}$
c, G, H	c/H	$c^3/(GH)$	H^{-1}
$\hbar, c, H$	c/H	$\hbar H/c^2$	H^{-1}
$\hbar, G, H$	$(\hbar G/H^3)^{1/5}$	$(\hbar^3 H/G^2)^{1/5}$	H^{-1}

The table is useful and cannot be complete. If Q_0 has the dimensions of a length, mass or time, then

$$Q = Q_0\, \Xi_H^s$$

has the same dimensions for every real s; allowing a general dimensionless coefficient gives $Q = Q_0 F(\Xi_H)$. The four displayed triples fix the otherwise free exponent by setting one constant's power to zero. Multiplication by Ξ_H^s turns their apparent finite menu into a continuum.

Nor does this remainder depend on retaining the pre-relativistic independence of L and T. In units with $c = 1$, time and length share a dimension and

$$[\hbar] = ML, \qquad [G] = M^{-1}L, \qquad [H] = L^{-1}.$$

Three quantities now span rank two, again leaving one dimensionless product: $\hbar G H^2$, the $c = 1$ form of Ξ_H. The remainder survives the change of convention that the historical box invites us to test.

3 A second remainder: Λ

Write the Planck and Hubble times as

$$t_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^5}}, \qquad t_H = H^{-1}.$$

Then

$$\Xi_H = \left(\frac{t_{\mathrm{P}}}{t_H}\right)^2 \sim 10^{-122}$$

at the present epoch. Equivalently, with $\rho_{\mathrm{P}} = c^5/(\hbar G^2)$ and $\rho_c = 3H^2/(8\pi G)$,

$$\frac{\rho_c}{\rho_{\mathrm{P}}} = \frac{3}{8\pi}\,\Xi_H.$$

The present-epoch ratio varies with H and differs from the constant usually invoked in the cosmological-constant problem. Introduce Λ, with $[\Lambda] = L^{-2}$. If Λ is treated as constant in the chosen expanding-frame description, the corresponding Buckingham–Bridgman group is

$$\Xi_\Lambda \equiv \frac{\hbar G \Lambda}{c^3} = \ell_{\mathrm{P}}^2 \Lambda \sim 10^{-122}.$$

The five quantities $\{\hbar, c, G, H, \Lambda\}$ still span rank three, so two independent remainders survive. Take

$$\Xi_H \equiv \frac{\hbar G H^2}{c^5}, \qquad \Xi_\Lambda \equiv \frac{\hbar G \Lambda}{c^3}.$$

Their ratio is

$$\frac{\Xi_\Lambda}{\Xi_H} = \frac{\Lambda c^2}{H^2}.$$

It is of order unity today (equal to $3\Omega_\Lambda$ in spatially flat ΛCDM). In that description, the small value of Ξ_Λ states the observed side of the vacuum hierarchy; the usual cosmological-constant problem also contains a cutoff-dependent field-theoretic estimate of vacuum energy. The coincidence problem appears as the present near-equality of two independent remainders. Dimensions explain neither fact.

> The observed dimensionless side of the cosmological-constant discrepancy is a Buckingham–Bridgman remainder. The field-theoretic estimate required to form the full discrepancy lies beyond dimensional analysis.

This also identifies the temptation worth resisting. The combination

$$m_W = \left(\frac{\hbar^2 H}{Gc}\right)^{1/3}, \qquad m_W^3 = m_{\mathrm{P}}^2 \left(\frac{\hbar H}{c^2}\right),$$

lies surprisingly near a hadronic, often pion-associated, mass scale. This Weinberg coincidence is not made explanatory by dimensional admissibility. In the presence of Ξ_H, multiplication by arbitrary powers of the hierarchy can manufacture many intermediate scales. A numerical resemblance becomes physics only when a mechanism selects it.

4 When the physical ruler changes

Wetterich's alternative description of cosmological redshift sharpens the point. The observation is a dimensionless frequency or wavelength ratio. In one field frame it is described mainly by an expanding scale factor; in another, particle masses grow and atomic rulers shrink. Since the Bohr scale behaves schematically as

$$a_B \sim \frac{\hbar}{\alpha m_e c},$$

an increasing electron mass shortens the material standard. "Expanding space" and "shrinking atoms" can therefore allocate the same dimensionless observation to different dimensioned quantities.[5, 6]

Fix cosmic time t for $g_{\mu\nu}$, and let

$$\widetilde{g}_{\mu\nu} = \Omega^2(t) g_{\mu\nu}, \qquad \mathrm{d}\widetilde{t} = \Omega\,\mathrm{d}t, \qquad \widetilde{a} = \Omega a.$$

With $H = \dot{a}/a$, where the dot denotes differentiation with respect to the original cosmic time, direct differentiation gives

$$\boxed{\widetilde{H} = \frac{1}{\Omega}\left(H + \frac{\dot{\Omega}}{\Omega}\right)}.$$

Thus H, and consequently Ξ_H, is not by itself a frame-invariant observable. The same audit applies to Ξ_Λ. In a variable-gravity frame the term playing the role of a cosmological constant and the gravitational coupling G can transform; the Planck scale constructed from G, $\hbar$ and c then transforms with them. Calling Ξ_Λ time-independent therefore records the assumption that Λ is constant in the chosen expanding-frame description. Buckingham's theorem establishes only that both remainders are invariant under passive rescaling of units. Their cosmic evolution and their status under field redefinition must be established from the model and from operational ratios.

Uzan's diagram already contains the bridge:

$$\alpha_G = \frac{G m_p^2}{\hbar c} = \left(\frac{m_p}{m_{\mathrm{P}}}\right)^2, \qquad \Xi_H = \left(\frac{m_H}{m_{\mathrm{P}}}\right)^2, \qquad m_H \equiv \frac{\hbar H}{c^2}.$$

Hence

$$\frac{\alpha_G}{\Xi_H} = \left(\frac{m_p c^2}{\hbar H}\right)^2.$$

In a variable-gravity description, the proton mass measured against the Planck mass is invariant under the conformal field redefinition while remaining free to vary with cosmic time. Those are distinct claims. This frame-invariant, time-dependent α_G is the handle on what changes. The Eötvös balance meets Ξ_H only indirectly, through the question of how material masses realise dimensionless gravitational strength.

5 The ruler becomes a test body

If a scalar field ϕ changes masses or couplings, a body A may be assigned a scalar charge schematically by

$$q_A = \frac{\partial \ln m_A(\phi)}{\partial \phi}.$$

The mass of a composite body includes nucleon masses, electromagnetic contributions and nuclear binding. Different materials need not have the same q_A. In a broad class of models the differential acceleration of bodies A and B toward a source S takes the form

$$\eta_{AB} \equiv \frac{2(a_A - a_B)}{a_A + a_B} \;\propto\; (q_A - q_B)q_S,$$

where the proportionality is model-dependent. A varying dimensionless coupling does not logically require composition-dependent fall: universal coupling is a special possibility. But non-universal coupling makes the equivalence test unavoidable.

The old separated-cube thought experiment belongs here only after the distinction is made. Tidal convergence asks how different points of an extended body sample a non-uniform field. An Eötvös comparison asks whether different internal compositions at the same event carry the same gravitational response. The two calculations expose different failures of the ideal "structureless point particle": spatial extent produces tidal convergence; an internal energy ledger can produce composition-dependent acceleration.[7]

The strongest tests compare just such ledgers. The final MICROSCOPE analysis compared titanium and platinum test masses in orbit and obtained, for its principal result,

$$\eta(\mathrm{Ti}, \mathrm{Pt}) = [-1.5 \pm 2.3\,(\mathrm{stat}) \pm 1.5\,(\mathrm{syst})] \times 10^{-15},$$

consistent with universal free fall at that precision.[8] The result does not constrain the numerical value of Ξ_H directly. It constrains theories in which the physics represented by changing dimensionless ratios couples differently to the constituents of the two bodies.

6 Matter that did not make the ruler

For ordinary laboratory matter, the answer has survived astonishing scrutiny. Yet the gravitating matter used in determining the cosmic history is not exhausted by beryllium, titanium, platinum, atoms or lunar rock. Some of it is known principally through gravitational inference and has never supplied a laboratory ruler or a test mass in an equivalence experiment. Nothing in dimensional analysis alone proves that this unseen sector carries the same scalar charge, the same operational α_G, or the same free fall as the matter from which our standards are made.

The question has already been partly asked. Torsion-balance experiments have compared the acceleration of ordinary materials toward the Galactic dark-matter distribution, and lunar laser ranging constrains differential Earth–Moon acceleration in external fields. Astrophysical methods extend the audit through galaxy clustering and other large-scale tracers of dark-matter motion.[9, 10, 11, 12] These are indirect, model-dependent tests of an unseen sector, rather than drops of an isolated dark-matter sample. Their existence sharpens the remaining question: how closely has each proposed dark sector been required to share the free fall of the matter that made our rulers?

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